Precept 6: Probability Exercises

October 27, 2026

Problem

The table below gives $P(x, y)$.

		x = 1	2	3
y	1	0.02	0.06	0.04
	2	0.10	0.36	0.22
	3	0.04	0.10	0.06

1. Verify this is a proper joint distribution.
2. What is the marginal $P(x)$?
3. What is the marginal $P(y)$?
4. What is the conditional $P(y \mid x = 1)$?
5. What is the conditional $P(x \mid y < 3)$?
6. Compute $\mathbb{E}[x] = \sum_{x=1}^{3} x\, P(x)$. What is this?

Solution

Question 1

To get a proper joint distribution, we need two things:

1. All entries must be non-negative,
2. All entries must sum to 1.

Indeed, all entries are non-negative. To show they sum to one, we can write

$$\underbrace{(0.02 + 0.06 + 0.04)}_{0.12} + \underbrace{(0.10 + 0.36 + 0.22)}_{0.68} + \underbrace{(0.04 + 0.10 + 0.06)}_{0.20} = 1$$

Question 2

To get the marginal $P(x)$, we sum out y, meaning that we sum down each column.

$$P(x = 1) = 0.02 + 0.10 + 0.04 = 0.16$$

$$P(x = 2) = 0.06 + 0.36 + 0.10 = 0.52$$

$$P(x = 3) = 0.04 + 0.22 + 0.06 = 0.32$$

As a sanity check, $0.16 + 0.52 + 0.32 = 1$.

Question 3

To get the marginal $P(y)$, we sum out x by summing across each row.

$$P(y = 1) = 0.02 + 0.06 + 0.04 = 0.12$$

$$P(x = 2) = 0.10 + 0.36 + 0.22 = 0.68$$

$$P(x = 3) = 0.04 + 0.10 + 0.06 = 0.20$$

As a sanity check, $0.12 + 0.68 + 0.20 = 1$.

Question 4

The conditional distribution is

$$P(y \mid x = 1) = \frac{P(x = 1, y)}{P(x = 1)}$$

Using $P(x = 1) = 0.16$, we can write

$$P(y = 1 \mid x = 1) = \frac{0.02}{0.16} = 0.125, \quad P(y = 2 \mid x = 1) = \frac{0.10}{0.16} = 0.625, \quad P(y = 3 \mid x = 1) = \frac{0.04}{0.16} = 0.25$$

which sums to 1, as every distribution should.

Question 5

The conditional distribution is

$$P(x \mid y < 3) = \frac{P(x, y < 3)}{P(y < 3)}$$

First, let's compute $P(y < 3)$. This is just summing from the marginal distribution

$$P(y < 3) = P(y = 1) + P(y = 2) = 0.12 + 0.68 = 0.80$$

Now, we can compute the following

$$\begin{aligned} P(x = 1 \mid y < 3) &= \frac{0.02 + 0.10}{0.80} = 0.15, \\ P(x = 2 \mid y < 3) &= \frac{0.06 + 0.36}{0.80} = 0.525, \\ P(x = 3 \mid y < 3) &= \frac{0.04 + 0.22}{0.80} = 0.325 \end{aligned}$$

Question 6

$$\mathbb{E}[x] = 1(0.16) + 2(0.52) + 3(0.32) = 0.16 + 1.04 + 0.96 = 2.16$$

This is the mean!